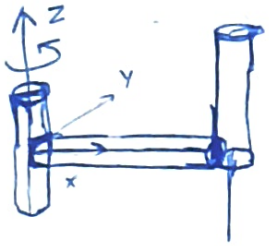
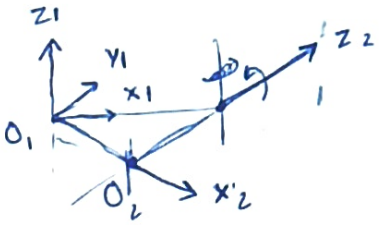
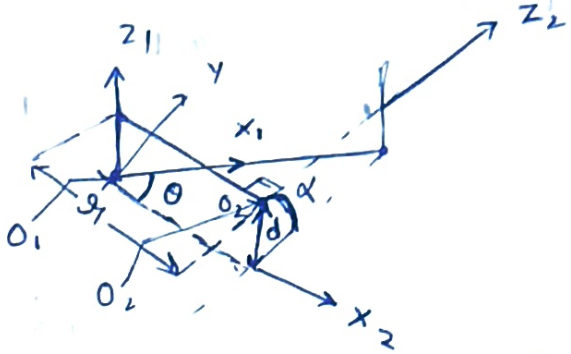
For first joint, z axis is axis of revolution
(axis of translation for prismatic joint)

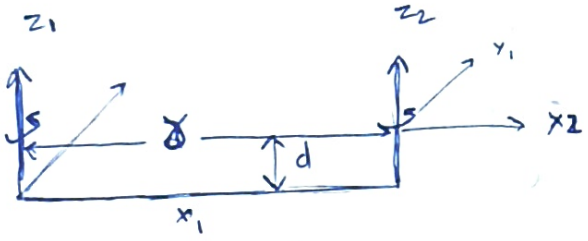
x-axis is free to be chosen (preferably in dir^n of link)
y-axis such that it completes RH co-ord system







Special case • $z_1 \parallel z_2$



$\alpha = 0$
r = same for all d
d = any distance
θ_1 = depends in current case $x_1 \parallel x_2$
$\Rightarrow \theta = 0$

r = length of common normal (also dir^n of x_2 is along normal)
d = (vertical) distance between O_1 & O_2 along z_1
θ = angle to rotate x_1 to bring it about x_2 / normal (about z_1)
α = rotation of z_1 about x_2 to align with z_2

Above was learned from youtube. Chatgpt gave the notation system as:

DH parameters: $\begin{bmatrix} a_i & \alpha_i & d_i & \theta_i \end{bmatrix}$

for pair of consequtive axis : $\underbrace{(x_{i-1}, y_{i-1}, z_{i-1})}_{\text{previous axis}}$ & $\underbrace{(x_i, y_i, z_i)}_{\text{current axis}}$

Revolute joint : θ_i variable
prismatic joint : d_i variable

$$^{i-1}T_i = R_z(\theta_i)\, T_z(d_i)\, T_x(a_i)\, R_x(\alpha_i)$$

i) Rotate around old z-axis by θ_i
ii) Translate along that z-axis by d_i
iii) Translate along new-x axis by a_i
iv) Rotate along that x-axis by α_i

$$R_z(\theta_i) = \begin{bmatrix} \cos\theta & -\sin\theta & 0 & 0 \\ \sin\theta & \cos\theta & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad T_x(a_i) = \begin{bmatrix} 1 & 0 & 0 & a_i \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$T_z(d_i) = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & d \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad R_x(\alpha_i) = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & C_\theta & -S_\theta & 0 \\ 0 & S_\theta & C_\theta & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Multiplying all four gives

$$^{i-1}T_i = \begin{bmatrix} C_\theta & -S_\theta C_\alpha & S_\theta S_\alpha & a C_\theta \\ S_\theta & C_\theta S_\alpha & -C_\theta S_\alpha & a S_\theta \\ 0 & S_\alpha & C_\alpha & d \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Important to note

End-Effector is $(i+1)$ when i is final revolute joint

so effectively for 2-DOF manipulator

$$T = {}^0T_2 = {}^0T_1 \, {}^1T_2$$

For ${}^1T_2$ / $1 \rightarrow 2$,

freedom to choose axis as no joint on 2, so preferably choose:

- $Z_2 \parallel Z_1$ at EE position
  X_2 along the link length

$\Rightarrow$ $d_2 = 0$

$a_2 = L_2$

$\alpha_2 = 0$

$\theta_2 = q_2$